Testing 3D games without SDK access

A global streaming platform offers cloud gaming, including a variety of casual games. Third parties submit the games, and each one must be verified before it is published. The platform provider wanted AI agents to perform autonomous testing, without SDK or codebase access, on released games to evaluate feasibility.

THE GOAL

Verify the game from a real user's perspective: played on a TV or browser screen, using a phone as a controller. Identify the limits of using AI agents to play without developer access.

Company
A global streaming platform

Industry
Cloud Gaming

Model
Games on a TV or browser window, played with phones as controllers

Testing types
Functional, Agent based

Pilot
Six weeks with 72 checks in scope

THE CHALLENGE AND HOW WE SOLVED IT

Aim and shoot within a timebox

A camera model turns the screenshot into a top-down grid, and a local model tests thousands of shots in seconds.

Reading a 3D course from a moving camera

A visual model (MobileSAM) outlines the course and walls, and image analysis finds the ball and the cup.

Working without code, SDK or documented controls

The agent plays through the screen and a phone controller, and the simulation shows what each action causes.

OUR APPROACH

1. Agree what can be verified

We explained which checks could not be met as written and agreed a deliverable version.

2. Track everything

A tracker marks each check proven, possibly flaky or to do, and doubles as the status report.

3. Screenshot to simulation

The agent finds the ball, target and walls, then simulates thousands of shots.

4. Run the heavy work locally

A local model returns the best power and direction in under five seconds.

5. Recreate the course in 3D

AI rebuilt the course as a Unity model, so simulations run in an accurate copy.

6. Orchestrate multiplayer

A supervisor agent coordinates player agents on remote machines.

24,000 shots from one screenshot

The agent finds the ball and the target, outlines the course, lays a grid of cells over it, simulates every shot at once, then picks the best one and takes it.

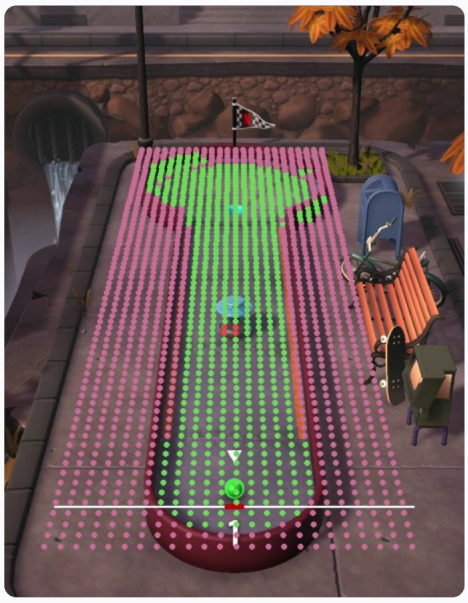


Fig. A
A grid of cells placed on the screen. Green cells fall inside the course outline.



Fig. B
The same roll put back on the screen: one bounce off the right rail, then into the cup.

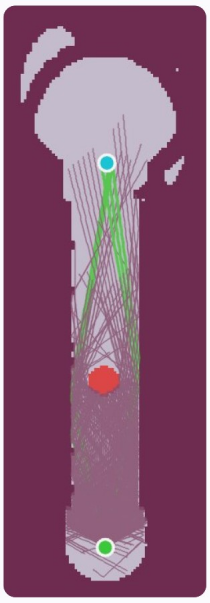


Fig. C
242 sample shots green hit the hole.

Model flexibility

Agentic QA works with cloud models and local models, or a mix. Here, a local model runs the simulations and returns the best power and direction.

3D copy

AI rebuilt the course as a Unity model, and simulations run in an accurate 3D copy.

Multiplayer

A supervisor agent coordinates player agents on remote machines, with actions landing within a few seconds.

24,000	5s or less	35	9 in 10
simulated shots per screenshot	to compute the best shots	holes sunk in live games	shots inside the 30-second timer

What worked

A screenshot became the foundation to enable agentic testing.

Where it applies

Black-box testing of games with no SDK or source.

Takeaway

The approach allows Agentic QA to cover similar games and challenges.